

Article

Special Education Status and Underidentification of Twice-Exceptional Students: Insights from ECLS-K Data

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Abstract: The current study examined the underidentification of students with disabilities for gifted education programs, otherwise referred to as twice-exceptional students. This study utilized data from the Early Childhood Longitudinal Study Kindergarten Class of 2010–2011 (ECLS-K). We estimated that approximately 17% to 18% more students with disabilities should have been identified for a gifted education program as having statistically similar achievement scores to those students without disabilities in gifted education programs. Alternatively stated, students with disabilities should make up 10.8% of gifted programs, or about 1 in 9 students in gifted programs should be twice-exceptional. Students with disabilities who were male, non-White, low-income, and indicating more internalizing problem behaviors were more likely to not be identified for a gifted education program despite having similar achievement scores.

Keywords: twice-exceptional; IEP; ECLS-K; underidentification; gifted



Citation: Jolly, J.L.; Barnard-Brak, L. Special Education Status and Underidentification of Twice-Exceptional Students: Insights from ECLS-K Data. *Educ. Sci.* **2024**, *14*, 1048. <https://doi.org/10.3390/educsci14101048>

Academic Editors: Michelle Ronksley-Pavia, Norma Hafenstein and Eleonoor van Gerven

Received: 15 August 2024

Revised: 16 September 2024

Accepted: 20 September 2024

Published: 26 September 2024



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1. Introduction

A discussion of twice-exceptional students is underrepresented in both the research literature and in gifted education programs. Consequently, the nature of these students' often unique learning profiles presents challenges in their identification for gifted and special education services [1]. The paucity of research on twice-exceptionality is partly attributed to the underidentification of this student population. In response, researchers have been responding to this discrepancy in the literature to aid practitioners in their identification of twice-exceptional students and the subsequent provision of appropriate programming and services. This growing corpus of the literature has become increasingly nuanced and includes the investigation of students with high ability and specific disabilities. These disabilities include specific learning disabilities (SLD), Attention Deficit Hyperactivity Disorder (ADHD), Autism Spectrum Disorder (ASD), and emotional disorders (ED) [2]. The two most common special education eligibility categories occurring with gifted identification include Other Health Impairment (OHI) and Specific Learning Disability (SLD) [3]. The nature of these students' distinctive learning profiles presents challenges in their identification for gifted and special education programming and services and is partly attributed to the underidentification of 2e students. Additional challenges include student learning profiles that present as typical but could be the result of compensatory abilities that mask a possible learning issue or disability. This study sought to shed light on the prevalence of underidentification using Early Childhood Longitudinal Study-Kindergarten (ECLS-K) data, specifically focused on IEP status and executive function and achievement scores.

1.1. Definitional Issues

A throughline in gifted and talented education, beginning with terminology and definitions, extending to identification, and culminating with programming and services, is essential. Confusion over these foundational elements can impact who is and is not

identified for gifted programs [4]. To complicate identification issues of 2e students, definitions are plentiful and can differ conceptually, depending on the theory of intelligence or model of giftedness and talent (see [5,6]). The absence of a universal definition adds to confusion around 2e identification [2,7,8]). Reis et al. (2014) [8] suggested the 14 disabilities (exclusive of intellectual disability) recognized by the U.S. Department of Education (disabilities categories: (a) autism, (b) deaf-blindness, (c) emotional disturbance, (d) hearing impairment, (e) intellectual disability, (f) multiple disabilities, (g) orthopedic impairment, (h) other health impairment, (i) specific learning disability, (j) speech or language impairment, (k) traumatic brain injury, (l) visual impairment, and (m) developmental delay) should be considered in a definition of 2e students and proposed an operational definition that outlined the potential for advanced achievement and the expression of one or more disabilities as detailed in federal or state special education requirements.

Provided with the various definitions of giftedness and twice-exceptionality and identification procedures used by school districts (and to a lesser degree states), the exact number of twice-exceptional students has also been difficult to determine. Estimates indicate that gifted students in the United States make up approximately 5% to 20% of the student population [2] of about 53 million students in grades K–12 [9], with the Office of Civil Rights (OCR) [10] reporting 6% of public school students enrolled in gifted and talented programs. In 2022–2023, the estimated number of students who received special education services was 7.5 million or 15% of the student population from ages 3–21 [9]. The number of twice-exceptional students was estimated to be around 360,000 students or 2% to 5% of gifted children [11,12], the same OCR reported 2.8% of gifted students with disabilities. The National Association for Gifted Education's 2020–2021 *State of the States Report* surveyed states regarding disaggregated gifted population data. Twenty-five states provided data for gifted students who received special education services, which ranged from 0.07 to 10 percent [13]. Cheek et al. (2023) reviewed the empirical literature on twice-exceptionality and developed a simulation study that found a prevalence rate of 0.14 in the most elastic conditions regarding definitions and inclusion criteria. The authors acknowledge that 2e students have unique learning needs. Still, improved adherence to the diagnostic criteria provides an empirically supported label and subsequent interventions and supports [4,14]. Further, estimates do not capture masking and compensatory behaviors, which may prohibit students' strengths and disability from being identified and supported [15].

1.2. Additional Considerations

Definition issues coupled with little to no pre-service education and in-service training regarding the educational needs of gifted and twice-exceptional students [2] create a fertile environment for underidentification. Most undergraduate teacher preparation programs do not cover gifted education in a meaningful way and twice-exceptionality is even less likely to be included. A handful of states require additional licensure to teach gifted and talented students and are typically part of graduate education programs [16]. In addition to a lack of preparation, teachers' misperceptions regarding twice-exceptional students are documented in the literature, including deficit beliefs for students who received special education services [17,18]. These educator-based factors can influence the under-identification of twice-exceptional students, which leads to their eventual underrepresentation in gifted and/or special education programs and appropriate educational interventions being offered [2,19]. Other school personnel, such as school psychologists, often play an integral role in the identification of students for special education services but are underutilized in the identification of twice-exceptional students [20]. Previous studies suggest that students with a disability are less likely to participate in programs for gifted and talented children, even when matching the characteristics and performance of gifted children without disabilities [11,12,17,21]. U.S. Public Law 94–142 or the Individuals with Disabilities Education Act [22] provides federal law and protection for the education and appropriate supports to students with disabilities. School and district characteristics also contribute to the rep-

resentation or underrepresentation of 2e students, which includes policies, availability of gifted programs, socioeconomic background of students, school size, and racial and ethnic makeup of the student population [23]. Yet, the identification and eventual participation in gifted and talented programs for these students suggested improved academic and affective outcomes for those students who are twice-exceptional [12].

Outside of disability status typically indicated by receiving special education services under the guidance of an Individualized Education Program (IEP), other variables have been associated with underidentification and underrepresentation in gifted education programs [24]. The contemporary literature has indicated that race/ethnicity has continued to be significantly associated with gifted education identification (e.g., [25–27]). Students who are African American have been less likely to be identified for gifted programs [28,29], and students who are Hispanic also have been less likely to be identified for gifted programs [30,31]. Additionally, research has indicated that students who were female have been less likely to be identified than children who were male [24,32,33]. These issues are certainly compounded with race such as in the case of Black boys who are less likely to be identified [10,34]. Students whose first language is not English also tend not to be identified [34]. Students from households with lower incomes have also been found to be less likely to be identified as gifted [35–37]. Students, who are older in age for their grade or were “redshirted” for kindergarten entry, have been positively associated with gifted identification [38]. Conversely, students who are younger in age for their grade have a higher likelihood of certain diagnoses such as ADHD, rather than being identified for gifted programming [39]. The intersectionality of these demographic factors only amplifies underrepresentation [24].

Beyond demographic variables, there has been less research examining how students identified for gifted education differ in terms of psychological factors and executive functioning skills, which also can prevent students from being identified, from students who are not identified. Bracken and Brown (2006) found that gifted students exhibited less internalizing and externalizing problem behaviors as well as fewer issues related to ADHD (i.e., attention problems, impulsivity, and lack of inhibitory control). Studies have found that gifted students appeared to have similar or better psychological well-being and life satisfaction than their peers [40,41]. These students typically do not exhibit school refusal attitudes. Kahn et al. (2014) suggested that gifted students typically do not exhibit school refusal attitudes or students who do not want to attend school regularly. The limited scholarship addressing gifted and twice-exceptional students’ school refusal is most often attributed to an academic and/or affective need not being addressed [42,43]. The literature also indicated school refusal attitudes among students with disabilities (e.g., [44,45]. For example, students with diagnoses of ADHD and ASD exhibited higher rates of school refusal due to bullying experiences at school [46].

1.3. Current Study

The purpose of the current study was to examine the degree of underidentification of students with IEPs for gifted programs and what factors were associated with this under-identification. We utilized data from a nationally representative sample of students from private and public schools across the United States via the Early Child Longitudinal Study-Kindergarten cohort of 2011 (ECLS-K:2011) [47].

First, we determined the percentage of students with IEPs who should be identified for gifted programs but were not identified when matched according to reading and mathematics achievement, respectively, as well as executive functioning. Gifted identification best practices recommend using multiple measures, which often include a measure of achievement [3,48]. We understand that gifted identification consists of more than achievement scores, but these scores provide an objective measure across a large, nationally representative sample.

Instead of a static cutoff, we utilized reading and mathematics scores as well as executive functioning (EF) scores to predict gifted program participation across students.

Barnard-Brak et al. (2015) [10] previously determined potential giftedness as being at the 90th or above percentile for both mathematics or reading achievement scores. Cain et al. (2019) also considered students scoring above the 90th percentile as gifted in their study of ASD and giftedness using the data from the Pre-Elementary Education Longitudinal Study (PEELS) and the Special Education Elementary Longitudinal Study (SEELS). While being in the top 10% has been used to determine potential giftedness, the implementation of this and other criteria can vary widely by jurisdiction (both local and state education agencies). Domain specific talent and achievement are important factors in gifted definitions [6] and models of giftedness and talent (e.g., [49,50]), which include an indication of high ability.

Executive function was included as a measure of cognitive ability when analyzed with math achievement scores and again with reading achievement scores. Previous research has suggested that EF, particularly working memory, can predict the advanced performance of young learners [51]. We calculated the posterior probabilities of each student being in a gifted program based on students with similar scores; thus, it was not dependent upon any percentile. Predictably, some students were correctly identified as being in a gifted program based upon these probabilities while others were not, who would have been under-identified according to achievement and executive functioning scores. Second, we examined the factors that contributed to any underidentification of students with IEPs and not identified as gifted but should have been based on achievement scores.

2. Method

2.1. Sample

The Early Childhood Longitudinal Study-Kindergarten (ECLS-K) [47] follows a large and nationally representative cohort of children beginning in 2010 across the United States, beginning in their kindergarten school year. The ECLS-K consisted of 18,174 children, which, when the appropriate weight was applied, equates to approximately 4,056,166 children across the United States. The ECLS-K: 2011 used a multistage sampling, stratified cluster design rather than a simple random sample to reduce field costs. First, the sample was divided into primary sample units (PSU) defined by counties or groups of neighboring counties. PSUs were stratified based on characteristics, including region of the country, median household income, and metropolitan status. Within the strata, PSUs were sampled based on probability proportional to size. Size was defined by the number of 5-year-olds in the PSU. The next sampling stage included selecting public and private schools providing kindergarten education within the sampled PSUs. Kindergarteners at each sampled school were selected [47].

Approximately 49% ($n = 8847$) were female and 51% ($n = 9288$) were male. With regard to race/ethnicity, approximately 47% ($n = 8489$) identified as White, non-Hispanic, 13% ($n = 2397$) as Black/African American, 25% ($n = 4690$) as Hispanic, 9% ($n = 1543$) as Asian, less than 1% ($n = 117$) as Native Hawaiian or Pacific Islander, 1% ($n = 168$) as American Indian or Alaska Native, and 5% ($n = 826$) as two or more races, non-Hispanic. Approximately 13% ($n = 1582$) had an Individualized Education Plan (IEP) on file with the school as of the spring semester of 3rd grade. We posited that most children would have been identified as having a disability by that age range. The average age in months was 108.98 ($SD = 4.41$) or 9.08 years. The age of 9 years was utilized as the Individuals with Disabilities with Education Act currently requires that children with developmental delay diagnoses be formally diagnosed or be declassified as special education [52]. Most gifted students are also identified by the 3rd grade for programming and services [53]. Before this age, children with a developmental delay diagnosis would have been included in the sample and may have subsequently been de-identified as special education.

2.2. Measures

All measures were obtained from the ECLS-K: 2011, which included the following assessments: (a) language screener, (b) reading (language and literacy), (c) mathematics, (d) executive function, (e) science, and (f) height and weight. For the purposes of this

study, the reading, mathematics, and executive function assessments were used. Executive functioning included measures of cognitive flexibility and working memory. The Dimensional Change Card Sort task assesses the cognitive flexibility of students [54]. The average score was 6.69 ($SD = 1.35$). The Numbers Reversed task assessed working memory and is part of the Woodcock–Johnson III Tests of Cognitive Abilities [55]. The average score was 95.83 ($SD = 17.04$) [47].

Approximately 7% ($n = 770$) were identified for a gifted program in reading by the spring semester of 3rd grade. Approximately 7% ($n = 775$) were identified for a gifted program in mathematics by the spring semester of 3rd grade. Among students with IEPs, approximately 4% ($n = 71$) were identified for a gifted program in reading and 5% ($n = 76$) in mathematics. Approximately 85% of the students with IEPs identified for a gifted program in reading were identified for a gifted program in mathematics as well. For both reading and mathematics achievement, Item Response Theory (IRT) scaled scores were utilized. A two-parameter logistic model for IRT-scaled scores incorporated both item discrimination and item difficulty in scoring [47].

Covariates of interest also included household income, age, sex, race, Hispanic status, inhibitory control, and teacher-reported conflict, as well as internalizing and externalizing problem behaviors and teacher-reported attention problems. Age was measured at the time of assessment in months. Inhibitory control was described as the “capacity to plan and to suppress inappropriate approach responses under instructions or in novel or uncertain situations” (Rothbart et al., 2001 [56], p. 1406). The average score was 3.69 ($SD = 0.84$). Internalizing problem behaviors typically refer to symptoms of depression and anxiety and are less likely to be observed. The average score was 1.58 ($SD = 0.52$). Externalizing problem behaviors typically refers to symptoms of aggression, hyperactivity, and impulsivity that would be generally observable. The average score was 1.71 ($SD = 0.61$). Teacher-reported attentional focusing was measured as the “capacity to maintain attentional focus on task-related channels. ‘When picking up toys or other jobs, usually keeps at the task until it’s done’”. [57] (p. 1406). The average score was 3.49 ($SD = 1.12$). For school attitudes, the average score was 7.42 ($SD = 1.97$).

2.3. Analyses

The ECLS-K has a complex survey design [47], whereby weights adjust for the differential or unequal probability of selection as well as design effects adjusted for the under-estimation of standard errors [58]. We applied these weights and design effects (e.g., cluster and stratification variables) in our analyses. In answering the research questions, logistic regression analyses were performed to predict who should have been identified for the gifted education programs in mathematics and reading, respectively, based on their respective academic achievements. In performing these analyses, model fit via a combination of pseudo- R -squared values, as well as the percentage of students who were correctly classified, was assessed. Values for the Nagelkerke and Cox and Snell pseudo r -squared were considered to be small, medium, and large for the respective values of 0.09, 0.15, and 0.25 or greater [56]. Given the complex survey design of the ECLS-K, only a handful of students per school (i.e., an average of 7.33 students per school) were represented across the 814 schools precluding multilevel [59] analyses with such small cluster sizes [60].

To answer the second research question, the posterior probabilities were utilized to classify children who should have been identified as gifted according to reading and mathematics, respectively. An estimate of the additional percentage of students with IEPs who should be identified for gifted programs in reading and mathematics, respectively, was provided. What variables are associated with these students who were matched according to achievement not being identified as gifted for reading and mathematics, respectively, via logistic regression analyses again were then examined.

3. Results

3.1. Predicting Gifted Program Participation

The model predicting gifted program identification based on reading achievement indicated an acceptable model fit with a Cox and Snell *R*-square value of 0.29 and a Nagelkerke *R*-square value of 0.11. Approximately 94.3% of students were correctly classified according to the regression model. Based upon the posterior probabilities (≥ 0.50), approximately 18% more students with IEPs should have been identified for participation in a gifted program when matched according to reading achievement. The model predicting gifted program identification based on mathematics achievement indicated acceptable model fit with a Cox and Snell *R*-square value of 0.33 and a Nagelkerke *R*-square of 0.11. Approximately 94.9% of students were correctly classified according to the regression model. Approximately 17% more students with IEPs should have been identified for participation in a gifted program when matched according to mathematics achievement. Approximately 85% of the students with IEPs identified for a gifted program in reading were identified for a gifted program in mathematics as well. Thus, we next examined what variables predicted students with IEPs not being in either reading or mathematics programs who should have based upon achievement scores.

3.2. Mathematics Achievement

The logistic regression results indicated acceptable model fit with a Cox and Snell *R*-square value of 0.06 and a Nagelkerke *R*-square of 0.47. Approximately 99.5% of students were correctly classified according to the model. Students from households with lower incomes were more likely not to be identified for a gifted program in mathematics versus students from households with higher incomes, $\beta = 0.08$, $SE = 0.01$, $p < 0.01$, $e^{\beta} = 1.17$. As negative attitudes toward school increased, the more likely not to be identified for a gifted program in mathematics, $\beta = 0.11$, $SE = 0.03$, $p < 0.01$, $e^{\beta} = 1.12$. As age increased, the more likely not to be identified for a gifted program in mathematics, $\beta = 0.07$, $SE = 0.01$, $p < 0.01$, $e^{\beta} = 1.07$. Additionally, students who exhibited more internalizing problem behaviors were more likely to not be identified for a gifted program when they should have been, $\beta = 0.31$, $SE = 0.13$, $p = 0.02$, $e^{\beta} = 1.37$. As the lack of inhibitory control increases, the more likely not to be identified for a gifted program in mathematics, $\beta = 0.33$, $SE = 0.12$, $p = 0.01$, $e^{\beta} = 1.40$. Students who were male were more likely not to have been identified as gifted when they should have been, $\beta = 0.70$, $SE = 0.1$, $p < 0.01$, $e^{\beta} = 2.01$. Students who were Hispanic were more likely not to have been identified as gifted, $\beta = 1.37$, $SE = 0.19$, $p < 0.01$, $e^{\beta} = 3.95$. Students who were African American were also more likely not to have been identified as gifted, $\beta = 1.50$, $SE = 0.38$, $p < 0.01$, $e^{\beta} = 4.49$. Students who were Asian were also more likely not to have been identified as gifted, $\beta = 1.40$, $SE = 0.27$, $p < 0.01$, $e^{\beta} = 4.07$. Finally, students with more attention problems were more likely not to have been identified as gifted, $\beta = 3.85$, $SE = 0.79$, $p < 0.01$, $e^{\beta} = 4.70$.

3.3. Reading Achievement

The logistic regression results indicated an acceptable model fit with a Cox and Snell *R*-square value of 0.06 and a Nagelkerke *R*-square of 0.35. Approximately 97.6% of students were correctly classified according to the model. Students from households with lower incomes were more likely not to be identified for a gifted program in reading versus students from households with higher incomes, $\beta = 0.36$, $SE = 0.03$, $p < 0.01$, $e^{\beta} = 1.43$. As negative attitudes toward school increased, the more likely not to be identified for a gifted program in reading, $\beta = 0.12$, $SE = 0.03$, $p < 0.01$, $e^{\beta} = 1.13$. As age increased, the more likely students were not to have been identified for a gifted program in reading, $\beta = 0.05$, $SE = 0.02$, $p < 0.01$, $e^{\beta} = 1.05$. Additionally, students who exhibited more internalizing problem behaviors were more likely not to be identified for a gifted program when they should have been, $\beta = 1.07$, $SE = 0.18$, $p < 0.01$, $e^{\beta} = 2.91$. As the lack of inhibitory control increased, the more likely students were not to have been identified for a gifted program in reading, $\beta = 0.41$, $SE = 0.14$, $p = 0.01$, $e^{\beta} = 1.51$. Students who were Hispanic were more

likely not to have been identified as gifted, $\beta = 2.54$, $SE = 0.32$, $p < 0.01$, $e^{\hat{\beta}} = 4.57$. Students who were African American were also more likely not to have been identified as gifted, $\beta = 0.60$, $SE = 0.29$, $p = 0.04$, $e^{\hat{\beta}} = 1.82$. Students who were Asian were also more likely not to have been identified as gifted, $\beta = 1.06$, $SE = 0.18$, $p < 0.01$, $e^{\hat{\beta}} = 2.90$. Finally, students with more attention problems were more likely not to have been identified as gifted, $\beta = 5.14$, $SE = 1.15$, $p < 0.01$, $e^{\hat{\beta}} = 3.76$. Table 1 provides the regression coefficients along with odds ratios ($e^{\hat{\beta}}$) for each covariate.

Table 1. Regression coefficients and odds ratios for covariates.

Covariates	Mathematics				Reading			
	β	S.E.	Sig.	$e^{\hat{\beta}}$	β	S.E.	Sig.	$e^{\hat{\beta}}$
Household Income	0.08	0.01	<0.01	1.17	0.36	0.03	<0.01	1.43
School Refusal Attitudes	0.11	0.03	<0.01	1.12	0.12	0.03	<0.01	1.13
Age at Assessment	0.07	0.01	<0.01	1.07	0.05	0.02	0.01	1.05
Internalizing Problem Behaviors	0.31	0.13	0.02	1.37	1.07	0.18	<0.01	2.91
Externalizing Problem Behaviors	0.28	0.16	0.08	1.33	0.35	0.18	0.05	1.42
Lack of Inhibitory Control	0.33	0.12	0.01	1.40	0.41	0.14	0.01	1.51
Sex	0.70	0.11	<0.01	2.01	−0.04	0.13	0.73	0.96
Hispanic Status	1.37	0.19	<0.01	3.95	2.54	0.32	<0.01	4.57
Race—White	0.26	0.30	0.40	1.30	5.27	0.43	0.10	1.93
Race—African American	1.50	0.38	<0.01	4.49	0.60	0.29	0.04	1.82
Race—Asian	1.40	0.27	<0.01	4.07	1.06	0.18	<0.01	2.90
Race—Native American	2.87	8.07	0.99	1.16	−0.69	8.01	0.93	0.50
Attention problems	3.85	0.79	<0.01	4.70	5.14	1.15	<0.01	3.76
Private School	1.70	0.30	<0.01	5.48	1.12	0.35	<0.01	3.35
Creative in work or play	−1.28	0.17	0.28	1.01	1.09	0.19	0.66	0.92
Lack of Gifted/Talented Teacher	0.53	0.21	<0.01	1.70	2.61	0.26	<0.01	13.59

4. Discussion

This study examined the underidentification of potentially gifted students with IEPs using mathematics and reading achievement scores as reported in ECLS-K (2010–2011 cohort) data. Achievement scores have often been used as the criterion for identifying students for gifted programs [61] (Kaufman & Sternberg, 2008). Findings from this study reaffirm results from previous research that twice-exceptional students have been under-identified based on their special education status [11,62]. We regressed reading and mathematics achievement scores, respectively, on gifted program participation for mathematics and reading. We then utilized the posterior probabilities to match which students should have been identified for a gifted program with respect to reading or mathematics scores.

4.1. Degree of Identification

We found that 18% more students with disabilities should have been identified for a gifted program, according to the reading achievement and executive functioning scores that were used as measures of cognitive function. For mathematics achievement and executive functioning scores (again used as a measure of cognitive function), we found that 17% more students with disabilities should have been identified for a gifted program. These students with disabilities (i.e., IEPs) who should have been identified for a gifted program have statistically similar levels of achievement in reading and mathematics as compared to students who were in gifted programs based upon posterior probabilities. These results quantify the degree of underidentification of twice-exceptional students relative to those students who had already been identified for a gifted program, rather than a static cutoff criteria such as a certain percentile score. Our study supports Cain et al.'s (2019) earlier findings that most students who scored at 90% or above on the Woodcock–Johnson Tests of Achievement–III [55] in their sample of PEELS data were not identified for gifted and talented programming. In addition, EF as a measure of cognitive function has been evidenced as a predictor of learning and achievement [63], including advanced

achievement [51]. The inclusion of EF with math and reading scores helped to establish the percentage of students who were under-identified in these two subjects.

The use of the EF score is not common in gifted education identification, even if a working memory score is provided with a measure of cognitive abilities (e.g., Woodcock–Johnson-III). Data-based decision making drives the review of students for special education eligibility. This team reviews a student’s response to intervention and progress monitoring data and evaluates a student’s progress using frequently collected data [22,64]. These data often include measures of cognitive ability and achievement. A gifted education specialist is typically not a member of this team, which results in little collaboration between special and gifted education specialists who could inform the data review and decision making. Thus, an opportunity for a student to be identified as twice-exceptional during this process is lost.

In addition, the underidentification of twice-exceptional students is not surprising based on several key issues identified in the literature. This unfamiliarity is reflected in who teachers refer for gifted education programs and how they complete teacher rating scales. Teachers’ deficit thinking can impact their conceptualization of students needing both special and gifted education services [65]. Additional factors include the lack of pre-service teacher education and in-service teacher professional learning regarding the needs of gifted and twice-exceptional students, which is rife in the literature [24]. This focus on a student’s challenges or deficits keeps educators from further investigating a student’s obscured strengths [15].

4.2. Factors Contributing to Underidentification

The covariates examined as factors contributing to underidentification are consistent with the previous literature, students from households with lower incomes were more likely not to be identified for a gifted program versus students from households with higher incomes. As negative attitudes by the student toward school increased, the student was more likely not to be identified for a gifted program in reading or mathematics when they should have been identified. As the lack of inhibitory control increased, the student was more likely not to be identified for a gifted program in mathematics or reading when they should have been identified. Also consistent with previous research, students who were Hispanic, Latinx or African American were also more likely not to have been identified as gifted in either reading or mathematics. Finally, students with more attention problems were more likely not to have been identified as gifted when they should have been.

Masking behaviors, however, such as emotional and behavioral issues often evidenced by twice-exceptional students also did not seem to be at issue [66–68]. Given the reported achievement scores were matched according to the posterior probabilities from the achievement model, any masking behaviors present among these twice-exceptional students were not substantial enough. Our results found that students who were male, non-White, low-income, and/or those who reported internalizing behavior problems were more likely not to be identified for a gifted program despite similar achievement scores. These findings are not dissimilar to earlier studies that also identified similar factors for non-identification for gifted programs, except internalizing behaviors [28,69–71].

Discussion has taken place regarding the internalizing behaviors of already-identified twice-exceptional individuals [72–74]; however, this study revealed a finding unique to students with disabilities. Those exhibiting internalizing problem behaviors were associated with a lower likelihood of being identified for a gifted education program in reading or mathematics when they should have been. Internalizing problem behaviors such as depression and anxiety may be considered as taking precedence over a student’s achievement as being more serious or severe. Externalizing problem behaviors such as hyperactivity or being impulsive may be considered a sign that a student is bored, and, therefore, not being challenged. This behavior may also indicate that further assessment may be required to determine if the attentiveness is linked to ADHD. This study does not account for stu-

dents who are underachieving academically, which could be linked to ADHD, boredom, environment, motivational factors, and/or self-efficacy [71].

Further, if the twice-exceptional research literature is scant, research regarding subpopulations within this group is even smaller. In our study, Asian students were also more likely not to have been identified as gifted in either mathematics or reading when they should have been. As Asian students are noted to be overrepresented in many gifted programs, receiving special education services could serve as an outsized influence when considering a student for referral or completing a teacher checklist as part of the gifted identification process. In a qualitative examination of Asian American parents of 2e students, Park et al. (2018) [72] reported that some parents found special education services were more easily obtained than gifted services, even though Asian students remain underrepresented in special education.

5. Conclusions

The current study provides an examination of students with IEPs with similar levels of achievement to that of students without IEPs in gifted education programs. Large, longitudinal samples such as the ECLS-K data set present the opportunity to examine the development of a community-based sample of students that would be nationally representative. The longitudinal nature of the data set provides more than a snapshot of the developmental course of students. Results indicate that approximately 17% to 18% more students with IEPs should be identified for gifted education programs (reading and mathematics, respectively). Alternatively stated, students with disabilities should make up 10.8% of gifted programs, or about 1 in 9 students in gifted programs should be twice-exceptional. Results also indicate that many of the demographic variables associated with underidentification of students without IEPs were associated with the underidentification of students with IEPs as well. Future research should further explore the phenomenon of students with more internalizing problem behaviors being under-identified for gifted education programs as compared to students with more externalizing problem behaviors.

Author Contributions: J.L.J. and L.B.-B. contributed equally to the development of the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: ECLS-K data are deemed exempt from review as a NCES data set per University of Alabama's Office for Research Ethics & Compliance.

Informed Consent Statement: Not applicable.

Data Availability Statement: The ECLS-K data set is a publically available data file (<https://nces.ed.gov/ecls/>) (access on 20 September 2024).

Conflicts of Interest: The authors declare no conflict of interest.

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